

## **TUJITUNZE**

### MVP Requirements Specification

#### *Functional & Non-Functional Requirements*

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## 1. Purpose

This document sets out the functional and non-functional requirements for the Tujitunze MVP: a member registers, links an account, and can then track every contribution and transaction in their own health savings wallet. It is the reference for what is being built and how it should behave.

## 2. Scope

Tujitunze's MVP centres on the member journey — registration, account setup, and full visibility into a personal health savings wallet — together with the platform operations needed to keep that wallet trustworthy: contribution rules, transaction processing, reconciliation, and audit logging.

## 3. Product Overview

Tujitunze is a digital health-savings and contribution platform for Tanzania. A member registers, links a mobile money account and, optionally, a bank account, and contributions are collected from their linked-account transactions according to configurable rules and credited to a personal health savings wallet, which the member can view, top up, withdraw from, and transfer between wallets at any time.

## 4. Users of the System

- **Member** — A registered individual using Tujitunze to manage their own health savings wallet.
- **Platform Operations (internal)** — Maintains contribution rules, processes linked-account transactions, and reviews reconciliation and audit records.

## 5. Functional Requirements

### 5.1 Registration & Account Setup

- **FR-1** — A new user can register as a member by providing full name, NIDA number, phone number, and a password.
- **FR-2** — The system verifies that the NIDA number and phone number are not already registered before creating an account.
- **FR-3** — Passwords are stored securely using industry-standard hashing, never in plain text.
- **FR-4** — During onboarding, a member selects their region and links a mobile money account as their primary contribution source.
- **FR-5** — Linking a bank account, during onboarding or later, is optional.
- **FR-6** — A registered member can log in with their NIDA number/email and password and receive a secure session.
- **FR-7** — A member can log out, and sessions expire automatically after a period of inactivity.

## 5.2 Profile & Account Management

- **FR-8** — A member can view their own profile: name, NIDA number, phone number(s), linked bank account (if any), and membership date.
- **FR-9** — A member can update their personal details and contact information.
- **FR-10** — A member can add additional phone numbers to their account.
- **FR-11** — A member can add, update, or remove a linked bank account at any time.
- **FR-12** — A member can change their password.
- **FR-13** — A member can reset forgotten password

## 5.3 Health Savings Wallet

- **FR-14** — A member can view their current wallet balance at any time.
- **FR-15** — A member can view a complete, itemised history of all wallet transactions — contributions, withdrawals, and transfers — each showing the balance before and after.

## 5.4 Contributions from Linked Accounts

- **FR-16** — Contributions are automatically calculated and credited to a member's wallet from their linked bank or telecom transactions, using configurable contribution rules (a fixed amount or a percentage of the transaction, with an optional minimum and maximum).
- **FR-17** — The same linked-account transaction is never processed into a duplicate contribution.
- **FR-18** — A member can view their telecom bundle, airtime, and payment history related to their contributions.

## 5.5 Notifications

- **FR-19** — A member receives a notification when their wallet is topped up, when a contribution is received, when a new phone number or bank account is linked, and when their password is changed.
- **FR-20** — A member can view and manage their notifications.

## 5.6 Platform Operations

- **FR-21** — Contribution rules (fixed or percentage-based, with source-type and amount limits) are configured and maintained centrally.
- **FR-22** — Every contribution and wallet-affecting action is recorded in an audit trail, capturing what happened, when, and for which member.
- **FR-23** — Linked-account transactions and the contributions generated from them can be reconciled to identify and resolve mismatches.

## 6. Non-Functional Requirements

- **Security & privacy** — NIDA numbers, phone numbers, and financial data are treated as sensitive. Every access to a member's data is authorised and scoped to that member. All input is validated on the server. Passwords are hashed with a strong algorithm. No secrets or credentials are stored in source code.
- **Data integrity** — Every wallet and contribution update happens as a single, all-or-nothing operation, so a failure never leaves a partial or inconsistent balance. Financial amounts are stored as precise decimal values, never floating-point numbers.
- **Reliability** — The contribution engine processes each linked-account transaction exactly once, even if it is received more than once.
- **Auditability** — Every sensitive action is logged at the moment it happens, so the audit trail always matches what actually occurred.
- **Performance** — A member's wallet balance and transaction history load in under half a second under normal conditions.
- **Availability** — The platform remains available and consistent even if a secondary process, such as logging, temporarily fails.
- **Usability** — A new member can register, view their wallet, and see their first contribution without needing help from support staff.
- **Transport security & abuse protection** — All communication with the platform is encrypted (HTTPS). Login and registration are protected against automated abuse through rate limiting.
- **Accounting model** — Wallet balances reflect Tujitunze's internal ledger of recorded contributions, top-ups, withdrawals, and transfers.

## 7. Data Foundations

The platform is built around the following core records:

- **Member accounts** — identity, contact, and credential information.
- **Contribution rules** — fixed or percentage-based rules that determine how much is contributed from a given transaction.
- **Linked-account transactions** — bank and telecom transactions recorded against a member's linked accounts.
- **Contributions** — the amount collected from a linked-account transaction and credited to a member's wallet.
- **Health wallets** — one wallet per member, holding the current balance.
- **Wallet transactions** — the full, itemised ledger behind every balance change.
- **Reconciliation records** — the match status between a linked-account transaction and the contribution it produced.
- **Audit trail** — an append-only record of every sensitive action taken on a member's contributions and wallet.

## 8. Assumptions

- Currency is Tanzanian Shillings (TZS).
- A member holds exactly one health savings wallet.
- Linked-account transaction data (bank or telecom) is received and processed to generate contributions.

## 9. Acceptance Criteria

The MVP is ready when the following hold true end-to-end:

1. A new member can register, log in, and view their own profile.
2. A linked-account transaction results in a correctly calculated contribution added to the member's wallet, even if it is received more than once.
3. The wallet balance and transaction history accurately reflect every contribution, withdrawal, and transfer.
4. A member can only ever see their own wallet, profile, and transaction history.
5. Every sensitive action is captured in the audit trail.

## 10. Glossary

- **Member** — a registered individual using Tujitunze.
- **Linked-account transaction** — a bank or telecom transaction recorded against a member's linked account, which may trigger a contribution.
- **Contribution** — the amount Tujitunze collects from a linked-account transaction and credits to a member's wallet, per an active contribution rule.
- **Health wallet** — a member's personal balance within Tujitunze.
- **Reconciliation** — comparing linked-account transactions against the contributions derived from them to catch mismatches.